

Capstone Design & Scoping

Overview

In AI engineering programs, the capstone is the integration point where learners prove they can deliver a full project under realistic constraints, not just train isolated models. Strong capstones combine technical depth with planning discipline, communication, and practical trade-off decisions.

What a capstone should demonstrate:

- end-to-end ownership from problem framing to delivery,
- evidence-based technical decisions,
- reproducibility and professional documentation,
- communication of impact, limitations, and risks.

How this curriculum maps into the capstone:

- Lessons 1-3: core programming, math, ML reasoning.
- Lessons 5-10: deep learning, GenAI, agentic systems, domain and edge contexts.
- Lessons 12-14: MLOps/LLMOps, safety, and frontier awareness.
- Lesson 11: product and research framing.

Lessons 1-14 Skill Traceability (Required)

Use this matrix as a mandatory coverage check before final capstone review:

Lesson	Skill signal expected in capstone evidence
1	clean code, modularity, testing, API hygiene
2	mathematically sound metrics, optimization/statistics reasoning
3	baseline modeling and rigorous model selection
4	deep learning architecture/training decisions (when relevant)
5	GenAI/LLM workflow design (prompting, RAG, generation quality controls)
6	foundational MLOps/LLMOps lifecycle and deployment discipline
7	agent workflow design, orchestration, and context handling
8	ethical risk framing, governance awareness, and responsible AI controls
9	specialization depth (RL/CV/NLP/domain-specific techniques) where applicable
10	edge/robotics constraints and system realism where applicable
11	product thinking, business impact framing, and research literacy
12	advanced observability, evaluation loops, and operational controls
13	safety, robustness, security, and guardrail strategies
14	frontier-awareness and evidence-based technical judgment

A capstone should not maximize complexity; it should maximize signal of competence.

Capstone Objectives & Competencies

Typical capstone objectives across AI engineering and university capstone programs:

1. define a real and bounded problem,
2. convert goals into measurable criteria,
3. execute a full AI lifecycle,
4. communicate decisions and outcomes professionally.

Core competencies expected in capstone syllabi and project-based programs:

- technical execution (data, modeling, evaluation),
- engineering rigor (versioning, testing, packaging, deployment),
- collaboration and team practices,
- oral/written communication,
- ethical and societal reflection.

Practical competency matrix:

Competency	Evidence in Capstone
Problem framing	Problem statement + stakeholder needs
Technical depth	Baselines, improved models, analysis
Engineering quality	Reproducible scripts, configs, deployable artifact
Professional practice	Status updates, decisions, presentation
Responsible AI	Risk/ethics section and mitigation notes

Project Selection & Scoping

Selecting a Capstone Topic

Evaluate candidate ideas on five dimensions:

- **impact**: meaningful business/research outcome,
- **feasibility**: can finish with available time/resources,
- **data readiness**: access and quality are realistic,
- **technical signal**: demonstrates relevant capabilities,
- **career alignment**: supports target role trajectory.

Statement of Work (SoW) Essentials

A strong SoW includes:

- project objective and non-objectives,
- concrete deliverables,
- assumptions and constraints,
- timeline and milestones,
- risk register with mitigation triggers,
- acceptance criteria.

Minimal SoW template:

1. Background and problem context.
2. Scope in/out boundaries.
3. Deliverables and quality bar.
4. Milestones and deadlines.
5. Risks, dependencies, and fallback plan.
6. Stakeholder review schedule.

Scope Control Pattern

Use a three-tier scope model:

- **Must-have:** minimum viable capstone.
- **Should-have:** high-value additions if schedule permits.
- **Could-have:** stretch goals only after core completion.

This prevents capstone failure from uncontrolled ambition.

Linking to Earlier Lessons

Choosing the Right Stack

Decide with role and problem fit, not hype:

- **Classical ML** when tabular prediction and interpretability dominate.
- **Deep learning** when unstructured data tasks require representation learning.
- **GenAI/RAG/Agents** when language reasoning, knowledge grounding, or workflow automation is central.
- **Edge/robotics** when latency/power/physical interaction constraints matter.
- **Research orientation** when novelty and hypothesis testing are primary outcomes.

Minimal Architecture Decision Questions

- What is the smallest system that proves value?
- Which lesson techniques are necessary vs optional?
- What can be simulated if production infra is unavailable?

Capstone Architecture Pre-Design Template

Before implementation, produce a one-page architecture sketch with these blocks:

1. Ingestion and data contracts.
2. Training/inference workflow (or RAG/agent workflow).
3. Evaluation harness and metric store.
4. Deployment target (batch, API, event-driven).
5. Observability hooks (logs, metrics, traces).
6. Safety/governance controls and escalation.

Text template:

```
User or data source -> data pipeline -> model/agent service -> decision/output layer
                    -> evaluation + monitoring store -> alerting/feedback loop
```

Design rule: If any block has no owner or no measurable success condition, scope is not ready.

Milestone Plan and Definition of Done

Recommended milestone cadence:

- **M1 (Week 1):** Problem framing, SoW, architecture draft, baseline KPI.
- **M2 (Week 2-3):** Data pipeline + baseline model/service.
- **M3 (Week 4-5):** Improved system + evaluation report + risk controls.
- **M4 (Week 6+):** Deployment/demo hardening + documentation + presentation pack.

Definition of done per milestone should include:

- objective evidence artifact,
- quality threshold,
- known risks and next actions.

Capstone Scoring Rubric (Detailed)

Use this rubric to prevent subjective grading:

Dimension	Weight	Evidence
Problem framing and scoping	20%	clear SoW, non-objectives, measurable success criteria
Architecture and implementation quality	25%	coherent design, modular code, justified trade-offs
Evaluation rigor	20%	baseline comparison, error analysis, limitations
Deployment/operations readiness	15%	runnable service or realistic deployment simulation + monitoring plan
Documentation and reproducibility	10%	setup/runbook, configs, artefact traceability
Communication and reflection	10%	presentation quality, Q&A clarity, lessons learned

Pass guidance:

- <70: incomplete lifecycle evidence.
- 70–84: solid execution with notable gaps.
- 85+: portfolio-ready capstone with credible end-to-end maturity.

Capstone Case Studies & Exceptions

Case 1: Enterprise-Style Product Capstone

A team built a document assistant with retrieval, API serving, and monitoring dashboard. They limited scope to one document domain and one primary workflow. Result: polished end-to-end demo and credible production discussion.

Case 2: Research-Style Capstone

A student compared causal vs correlational methods on a synthetic + real dataset and delivered strong ablation analysis, even without a production API. Result: high research signal with clear limitations.

Case 3: Domain-Focused Capstone

A healthcare-oriented project used conservative model choices and heavy documentation on ethics and risk handling. Result: less model novelty but stronger real-world viability.

When Smaller is Better

A focused, complete project often outperforms an over-scoped unfinished system. If time is limited, prioritize:

- one high-quality workflow,
- one robust evaluation story,
- one clear deployment or simulation artifact.

Interview Questions & Answers

1. **Q:** How would you scope an AI capstone project? **A:** Define problem, stakeholders, constraints, and success metrics; then reduce scope to a must-have MVP with milestone gates.

2. **Q:** What makes a capstone realistic? **A:** Feasible timeline, accessible data, measurable objective, and clear fallback options.
3. **Q:** How do you choose between ML and LLM approaches? **A:** By task fit, data type, explainability needs, cost, and delivery timeline.
4. **Q:** What is a statement of work in capstone context? **A:** A formal scope contract describing objectives, deliverables, milestones, risks, and acceptance criteria.
5. **Q:** What is a common capstone failure mode? **A:** Overly broad scope without milestone-based de-risking.
6. **Q:** How do you define capstone success? **A:** Technical metrics + delivery quality + reproducibility + stakeholder clarity.
7. **Q:** How should learners handle uncertain data availability? **A:** Add contingency datasets and synthetic prototypes early.
8. **Q:** Why include non-objectives? **A:** To prevent scope creep and expectation mismatch.
9. **Q:** What if the best model is not deployable in time? **A:** Deploy simpler robust baseline and document upgrade path.
10. **Q:** Why are communication competencies part of capstone grading? **A:** Real AI projects succeed through cross-functional alignment, not modeling alone.
11. **Q:** How do you pick an impressive but feasible capstone? **A:** Select a narrow real problem and deliver complete lifecycle evidence.
12. **Q:** Should capstones prioritize novelty? **A:** Only if baseline execution is still solid; novelty without delivery is weak.
13. **Q:** What is the role of risk register? **A:** Early identification of blockers and explicit mitigation planning.
14. **Q:** How do you align capstone with career goals? **A:** Match project artifacts to target role expectations (MLOps, GenAI, product, research).
15. **Q:** What is an acceptable minimum capstone artifact set? **A:** Reproducible code, evaluation report, architecture notes, and clear demo.
16. **Q:** Why should capstones include ethics/safety reflection? **A:** Responsible AI is expected in modern production and leadership contexts.
17. **Q:** How often should scope be revisited? **A:** Weekly, with explicit must/should/could re-prioritization.
18. **Q:** What is a capstone milestone gate? **A:** A checkpoint where continuation depends on objective completion criteria.
19. **Q:** What if team members have different priorities? **A:** Use decision logs and role clarity to resolve trade-offs transparently.
20. **Q:** What would you do in week 1 of a capstone? **A:** Finalize problem framing, SoW, success metrics, dataset plan, and risk assumptions.

Source-Informed References

- IBM AI Engineering Professional Certificate: <https://www.coursera.org/professional-certificates/ai-engineer>
- AI Capstone Project with Deep Learning (IBM/Coursera): <https://www.coursera.org/learn/ai-deep-learning-capstone>
- Monash FIT3193 AI Project (capstone learning outcomes): <https://handbook.monash.edu/2026/units/fit3193>
- Miami Dade College AI Capstone competencies: [https://www.mdc.edu/asa/documents/competencies/pdf/CAI4950C%](https://www.mdc.edu/asa/documents/competencies/pdf/CAI4950C%20AI%20Capstone%20Competencies.pdf)
- South Dakota AIE 465 AI Engineering Capstone II: <https://www.sdstate.edu/sites/default/files/file-archive/2026-04/AIE%20465%20Artificial%20Intelligence%20Engineering%20Capstone%20II%20%28NCR%29.pdf>